

Magneto-optic-based devices for polarization control

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Abstract

In this paper we present a number of measurements of the magneto-optic (MO) polarization rotation effect in MO glasses. We then show three design concepts of MO based devices that are of interest for optical communications applications. The first device is a tunable optical polarization controller, the second is a polarization filter that passes through only a specific point on the Poincaré sphere and the third is a MO Fabry–Perot resonator that enhances the MO effect and can be used inside an all-optical switch or as part of a polarization coding module.

Keywords: magneto-optical materials, all-optical devices, polarization

1. Introduction

The magneto-optical (MO) effect is used in a variety of applications in optics. The polarization rotation property and the non-linear magnetic response of magneto-optic materials make them attractive to use as optical isolators, modulators, etc [1–4].

In this paper we present a number of measurements of the magneto-optic polarization rotation effect in magneto-optic glasses. We show the difference between the ‘linear’ response and the hysteresis response of different materials.

We also present three design concepts for MO based devices that produce different types of polarization control. The first device is a tunable optical polarization controller (TOPC) that can convert an arbitrarily polarized input wave to any desired linearly polarized wave. This device is based on two feedback controlled MO rotators. The second device is an all-optical Poincaré dot polarizer (AOPDP). This device will pass through a wave only if it has a predefined polarization state that is represented by a dot or a small area on the Poincaré sphere. A wave that is differently polarized will be rejected and blocked. The third device is a MO Fabry–Perot (FP) resonator. We show that this resonator can produce enhancement of the MO effect by more than one order of magnitude. We also show the unique frequency dependence of the resonator’s transmission function.

Apart from the fact that all three devices are based on MO materials, they are also relatively simple to implement and therefore attractive for optics communication applications.

In section 2 we present the theory of the MO effect. Experimental measurements are presented in section 3. Sections 4–6 describe the three applications of tunable optical controller, polarization filter and MO resonator. The paper is concluded in section 7.

2. Magneto-optic effect—theory

The MO effect, known as the Faraday effect, is the creation of circular birefringence (CB) in a medium due to an induced magnetic field [4, 5]. We say that a material has CB when right circular polarization (RCP) and left circular polarization (LCP) experience a different refractive index. As shown by Faraday and many others later, there exists a wide group of materials that experience a significant MO effect. These are called magneto-optic materials.

In order to get CB in MO materials one usually (except when there is natural CB) needs to induce a magnetic field parallel to the wave’s direction of propagation. Let us denote the refraction index difference caused by the magnetic field as

$$\Delta n(\mathbf{B}) = n_+ - n_-, \quad (1)$$

where n_+, n_- are the refractive indexes for RCP and LCP respectively.

An important property of the MO effect is that it leads to polarization rotation of any arbitrary linearly polarized wave propagating in the CB medium. Suppose a wave is propagating in a CB medium. If the field at the entrance is $E = E_0 \hat{e}_x$ then

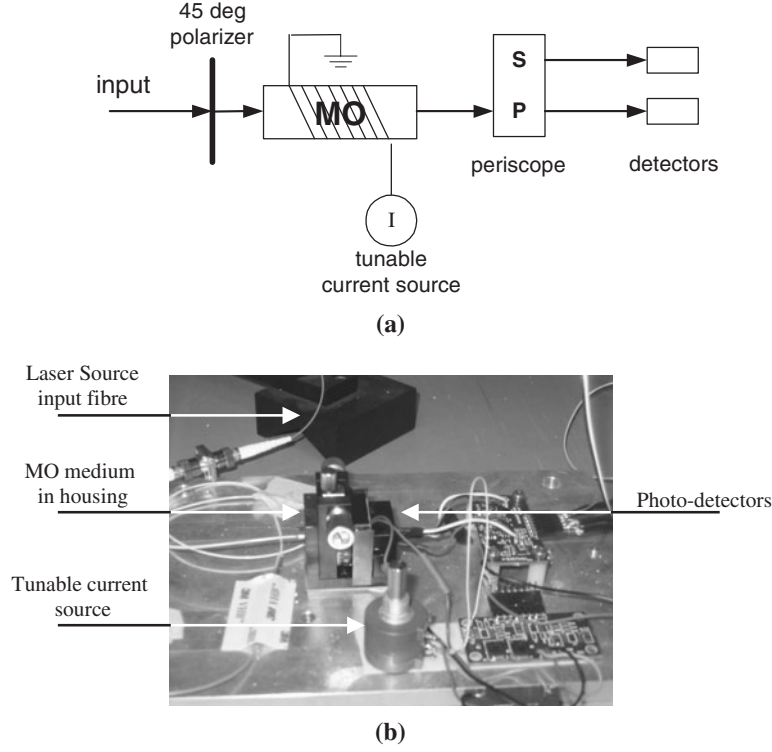


Figure 1. (a) Schematic illustration of the experiment set-up for measuring the polarization rotation as a function of the induced magnetic field. (b) Photograph of the actual experiment set-up.

the field after propagating a distance z will be

$$E = \frac{1}{2}E_0[(\hat{e}_x + j\hat{e}_y)e^{j(\omega t - 2\pi n_+ z/\lambda_0)} + (\hat{e}_x - j\hat{e}_y)e^{j(\omega t - 2\pi n_- z/\lambda_0)}]. \quad (2)$$

The last equation can be written in the form

$$E = \frac{1}{2}E_0 e^{j(\omega t - \pi \Delta n z/\lambda_0)} \left[\hat{e}_x \cos \frac{\delta}{2} + \hat{e}_y \sin \frac{\delta}{2} \right], \quad (3)$$

where

$$\delta = 2\pi \Delta n z/\lambda_0. \quad (4)$$

It is immediate from the last two equations that advancement in the CB medium leads to polarization rotation. When used to create polarization rotation the MO medium is often called a Faraday rotator. We should note that Δn will be positive or negative depending whether the direction of the induced magnetic field is with or against the wave's direction of propagation, respectively. This directional property of the MO effect makes Faraday rotators non-reciprocal devices and as such very useful for application requiring optical isolation.

The Faraday rotator can be represented by the following Jones and Mueller matrices [5]:

$$\mathbf{F}_{\text{Jones}} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}, \quad (5)$$

$$\mathbf{F}_{\text{Mueller}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\alpha & -\sin 2\alpha & 0 \\ 0 & \sin 2\alpha & \cos 2\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where α is the rotation angle.

It is important to note that the hysteresis phenomenon that occurs in the magnetization of ferromagnetic materials due to

magnetic domains also affects the MO properties. Therefore, in many MO materials we will witness saturation and hysteresis phenomena when measuring the MO effect in relation to the induced magnetic field.

3. Magneto-optic effect—measurements

We have performed several measurements on MO glasses that consist of rare earth iron garnet films grown epitaxially on GGG (gadolinium gallium garnet) substrates in order to explore their MO effect. We used different glasses¹ and measured the polarization rotation of a wave that propagates through the MO glasses as a function of the induced magnetic field. All MO glass substrates were manufactured with a thickness that creates 45° rotation for a 1550 nm wave when the MO effect is saturated.

The experiment set-up is described in figure 1. A collimated light at 1550 nm was passed through a polarizer of 45° (relative to the system's reference frame). The light coming from the polarizer was composed of two equally powered components that correspond to two orthogonal polarization states—P and S. After passing through the MO glass under test that was mounted in the centre of an electromagnet, the two states were separated by a periscope to the P and S states. After passing through P and S polarizers, the power of each wave was measured by two photodetectors. The rotation angle was calculated from the ratio of the normalized power of the P and S signals. By changing the current in the electromagnet we were able to control the magnetic flux

¹ The glasses are of the same type, but differently manufactured, and contain different concentrations of ingredients.

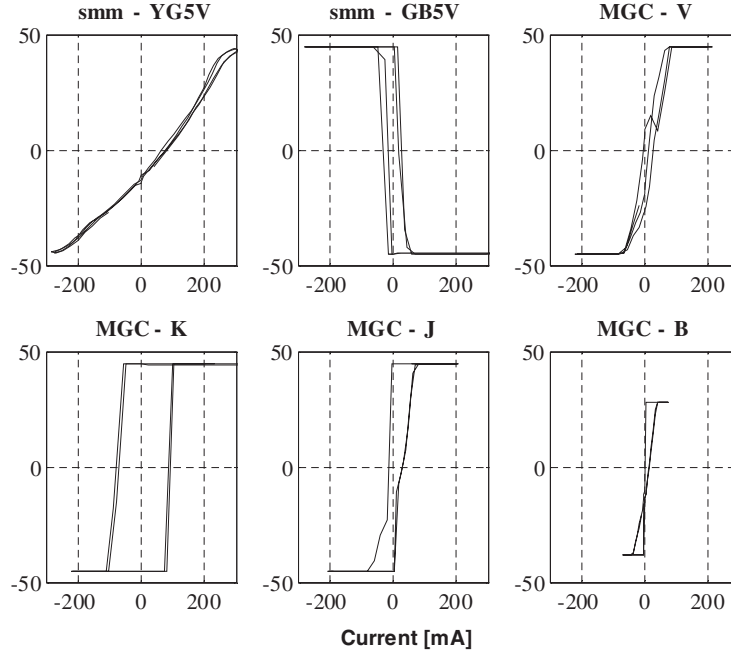


Figure 2. MO effect measurements results. Each graph represents a measurement of the polarization rotation angle (deg) versus the coil's current (mA), performed on a specific glass. The glasses are denoted by their manufacturer and part number (noted above each graph).

density which, as we explained earlier, changes the 'rotation velocity'. The coil used was made of 364 turns of 0.1 mm wire. The coil's inductance was 5.2 mH and its DC resistance was 17.5 Ω .

The rotation angle as a function of the current in the coil was measured for every sample. We measured from -45° rotation to $+45^\circ$.

Figure 2 describes the measurements results. For each glass we present two runs of measurements. The graphs depict the rotation angle as a function of the current in the coil.

Several important properties of the MO effect in the glasses can be derived from the experiment results that can be useful for different applications. One can see that all glasses experience the hysteresis phenomenon. This phenomenon can be characterized by several parameters. The most trivial parameter is the hysteresis maximum width. It can be seen in figure 2 that the hysteresis width is different for different types of glasses. In particular, for smm-YG5V the maximum width is small. Thus, this glass is referred to as linear, meaning that the hysteresis phenomenon is negligible. Such glass is suitable to be used as a polarization rotator. Glasses with large hysteresis width are more common in devices such as switches having variable optical attenuation or multicasting capabilities. Another parameter is the saturation field intensity. One can see that different glasses require different saturation field intensities. This parameter is of high importance for any application that uses MO glasses. Finally, one can also note that the glasses differ in their 'hysteresis linearity', i.e. the linearity of the function lines between the saturations points. The MGC-V glass, for example, has a highly non-linear hysteresis. Such glass cannot be used in applications that need to perform in the non-saturated area.

4. Tunable optical polarization controller

We will now describe a design concept for a TOPC that enables us to transform an arbitrary polarization to any desired linear polarization. The proposed TOPC active components are MO rotators, it has a very simple feedback control and it preserves all the energy of the incoming waves, unlike conventional polarization controllers.

We will first show the basic representation of wave polarization in order to establish common signs and concepts. Electromagnetic waves can be represented using their electric induction vector:

$$\mathbf{D} = (A_x e^{j\phi_x} \hat{x} + A_y e^{j\phi_y} \hat{y}) e^{j(\omega t - kz)}, \quad (6)$$

where $\hat{x}$, $\hat{y}$ are two orthogonal unit vectors, A_x , A_y and ϕ_x , ϕ_y represent the amplitudes and absolute phases respectively, ω is the wave's frequency, k is the wave number and z stands for the propagation distance.

In isotropic media, the polarization of the wave is parallel to $\mathbf{D}$. Thus, the temporal evolution of the polarization vector can be described by the evolution of $\mathbf{D}$. In the most general case, the rotation of the polarization vector (due to the harmonic nature of the wave) creates an ellipse in the $\mathbf{D}$ plane. Figure 3 depicts such an ellipse representing an arbitrary polarization. The ellipse's azimuth and ellipticity are determined by the wave's parameters— A_x , A_y , $\phi = (\phi_x - \phi_y)$. The ellipse parameters can be derived using [5]

$$\begin{cases} a^2 = A_x^2 \cos^2 \alpha + A_y^2 \sin^2 \alpha + 2A_x A_y \cos \alpha \sin \alpha \cos \phi \\ b^2 = A_x^2 \sin^2 \alpha + A_y^2 \cos^2 \alpha - 2A_x A_y \cos \alpha \sin \alpha \cos \phi \\ ab = \pm A_x A_y \sin \phi \end{cases} \quad (7)$$

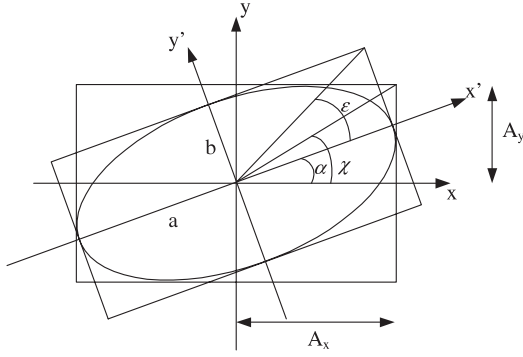


Figure 3. Ellipse description of the general state of polarization.

$$\begin{cases} \tan \varepsilon = \frac{b}{a} \\ \tan 2\alpha = \frac{2A_x A_y}{A_x^2 - A_y^2} \cos \phi. \end{cases} \quad (8)$$

The azimuth, α , and the ellipticity, ε , can be also represented graphically by the well known Poincaré sphere. The axes of the Poincaré sphere are given by the normalized Stokes parameters:

$$\mathbf{S}^{\text{in}} = \begin{cases} S_0^{\text{in}} = 1 \\ S_1^{\text{in}} = \frac{A_x^2 - A_y^2}{A_x^2 + A_y^2} \\ S_2^{\text{in}} = \frac{2A_x A_y \cos \phi_{\text{in}}}{A_x^2 + A_y^2} \\ S_3^{\text{in}} = \frac{2A_x A_y \sin \phi_{\text{in}}}{A_x^2 + A_y^2}. \end{cases} \quad (9)$$

Figure 4 shows a scheme of the proposed TOPC. The TOPC contains three electrically controlled MO rotators. Between the first two MO rotators a quarter-wave plate (QWP) is placed. The operating concept is illustrated in figure 5. An arbitrarily polarized wave is directed to the first MO rotator (MO1). The purpose of MO1 is to bring the wave's polarization to a point where $\alpha = 0$, i.e. the ellipse's major axis coincides with the $\hat{x}$ direction of the TOPC's reference frame. This is done using a feedback signal that is generated by a polarized beam splitter (PBS) placed after MO2. The PBS directs the $\hat{y}$ polarized part of the light to a detector. In order to understand how to use the feedback signal we shall calculate the light intensity at the detector. We use Jones calculus to obtain the field there. The field at the PBS is obtained by

$$\begin{pmatrix} D_x \\ D_y \end{pmatrix} = \begin{pmatrix} -j & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix} \begin{pmatrix} A_x e^{j\phi} \\ A_y \end{pmatrix}, \quad (10)$$

where A_x , A_y and ϕ represent the input wave. θ_1 is the rotation angle imposed by MO1 and the other constant matrix represents the QWP. MO2 does not create any rotation at this moment and therefore does not affect the propagating wave. Thus, the light intensity at the detector is $|D_y|^2$, given by

$$|D_y|^2 = A_x^2 \cos^2 \theta_1 + A_y^2 \sin^2 \theta_1 - A_x A_y \sin 2\theta_1 \cos \phi. \quad (11)$$

The local minimum and maximum points of $|D_y|^2$ can be found using the derivative with respect to θ_1 . The result is

$$\tan 2\theta_1^{\text{m}} = \frac{2A_x A_y}{A_y^2 - A_x^2} \cos \phi. \quad (12)$$

By using the second derivative it is easy to show that θ_1^{m} is a minimum point. Also from equations (8) and (12) it is clear that

$$\theta_1^{\text{m}} = -\alpha. \quad (13)$$

Therefore, when the intensity at the detector is minimal we know that the polarization azimuth of the wave coming out of MO1 equals zero. Thus, the feedback signal will command MO1 to rotate the polarization until the minimum reading is detected. Using the trivial algorithm of rotating the polarization towards the direction that reduces the detector's reading, finding the minimum point should not require more than 90° rotation.

After MO1 reaches steady state, i.e. the minimum point, the wave that enters the QWP has exactly 90° phase shift between its $\hat{x}$ and $\hat{y}$ components. The QWP then linearizes the wave's polarization by creating another phase shift of 90° . The purpose of the second MO rotator (MO2) is to rotate the linearly polarized wave emerging from the QWP so that the wave's polarization coincides with the $\hat{x}$ axis of the PBS. Once again we use the feedback signal that is produced by the $\hat{y}$ component of the polarization and the detector. In this case it commands MO2 to rotate the wave's polarization until reaching the minimum point. In this case, the minimum point should be zero intensity since the wave is linearly polarized.

Finally, the $\hat{x}$ polarized wave emerging from the PBS can be tuned to become any desired linear polarization using the third MO rotator (MO3).

To summarize, MO1 rotates the polarization ellipse so its major axis becomes parallel to the $\hat{x}$ direction. The action of MO1 combined with the QWP result in a linearly polarized wave at the exit of the QWP. MO2 rotates the wave's linear polarization so it coincides with the $\hat{x}$ direction of the PBS in which case all the wave's energy it passed to MO3. MO3 enables the operator to tune the polarization to any desired linear polarization.

It is important to note that throughout the analysis of the TOPC we assumed that only one wavelength is fed into the system. If we want the TOPC to be able to operate properly in a wide range of wavelengths we have to make sure that the QWP covers all of the desired wavelength range. Another option is to use an electro-optical (EO) modulator. An EO modulator can be used as a QWP for a range of wavelengths according to the static electric field that is induced in its medium. In that case the operator will need to adjust the field level according to the input wavelength. Also, we would have to make sure that the MO rotators are capable of performing $\pm 180^\circ$ rotation for the entire desired wavelength range.

5. All optical Poincaré dot polarizer

In this section we use the TOPC to suggest a design concept for an AOPDP. An AOPDP is a device that passes light through only if its polarization corresponds to a specific, pre-defined dot on the Poincaré sphere. All other polarizations are blocked.

The design we suggest for the AOPDP is described in figure 6. The AOPDP is based on two degenerate TOPCs (described in section 4), and a photo-refractive (PR) [7, 8] switch. As shown before, the TOPC is capable of transforming a wave's polarization to a linear polarization. In particular, we

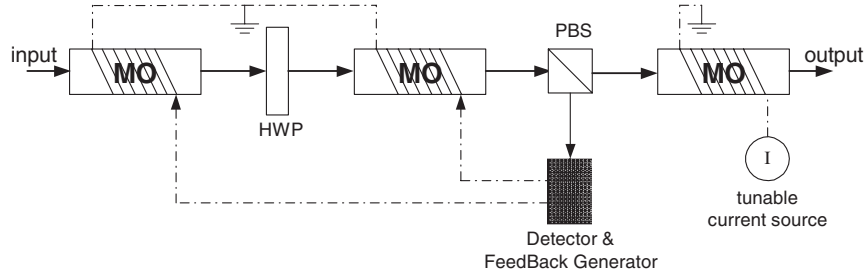


Figure 4. Scheme of the TOPC. Dashed lines represent electrical connections, solid lines represent light waves.

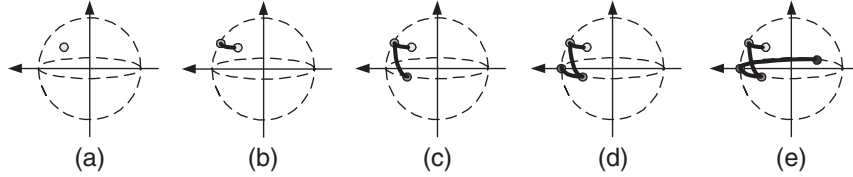


Figure 5. Illustration of the operating concept using the Poincaré sphere. The polarization state evolution described is (a) input stage; (b) after MO1; (c) after the QWP; (d) after MO2 and (e) after MO3.

can adjust the rotation angles of MO1 and MO2 to a fixed value, remove the feedback signals and remove MO3 so that only a uniquely polarized wave will be transformed completely into a $\hat{x}$ linearly polarized wave. If the input field is described by A_x , A_y and ϕ then the appropriate angles of MO1 and MO2 are given by

$$\begin{aligned} \tan 2\theta_1 &= \frac{2A_x A_y}{A_y^2 - A_x^2} \cos \phi \\ \tan \theta_2 &= j \frac{\exp(j\phi) A_x \cos \theta_1 + A_y \sin \theta_1}{-\exp(j\phi) A_x \sin \theta_1 + A_y \cos \theta_1}. \end{aligned} \quad (14)$$

The QWP needs to be positioned to delay the $\hat{x}$ polarization or the $\hat{y}$ polarization according to the sign of ϕ .

We should note that if a differently polarized wave is directed into the degenerate TOPC then part of the wave's power will be directed in the $\hat{y}$ direction of the PBS. This is true since all three components (MO1, MO2, QWP) are represented by one-to-one functions on the Poincaré sphere. Thus, the field coming out of the $\hat{y}$ direction of the PBS indicates the validity of the input wave's polarization.

The waves emerging from the $\hat{x}$ and $\hat{y}$ directions of the PBS are directed into two inputs of a PR switch. The PR switch can be designed so that if In2, of the PR switch, is present then the energy of In1 is coupled to the Out2 port, otherwise the energy of In1 will propagate to the Out1 port. The logic function of the switch is thus $\text{Out2} = \text{In2} \cdot \text{In1}$ and $\text{out1} = \overline{\text{In2}} \cdot \text{In1}$.

In the AOPDP device we direct the $\hat{x}$ polarized wave (coming out of the first degenerate TOPC) into the PR switch In1 and the $\hat{y}$ polarized wave into In2. Note that both waves are passed through an optical amplifier. We assume that the amplifiers are designed to work in saturation mode so that given that their input signal is sufficiently powerful their output will be constant and independent of its input. These conditions allow the proper functioning of the PR switch. In addition, figure 6 also presents an alternative implementation of the analyser section that contains the PR switch. In the alternative

implementation we use a PR wavelength converter [7]. The logical result is identical.

Thus, any wave that is not polarization-matched to the rotation angles of MO1 and MO2 will generate a wave in the $\hat{y}$ direction of the PBS and by that activate the PR switch, i.e. block the $\hat{x}$ polarized wave from propagating through. Only the polarization-matched wave (the Poincaré dot) will not activate the PR switch and will enable the propagation of the $\hat{x}$ polarized wave. In the latter case the wave propagating out of the PR switch is directed into an 'inverse degenerate TOPC'. This device is very similar to the first degenerate TOPC but does exactly the inverse procedure, i.e. it transforms the $\hat{x}$ polarization of the wave to be equal to the polarization of the input wave. To enable this situation the rotation angles of MO3 and MO4 should be

$$\theta_3 = -\theta_2$$

$$\theta_4 = -\theta_1$$

and the QWP should be rotated 90° in relation to the QWP in the previous TOPC.

To summarize, the combination of a degenerate TOPC followed by a PR switch followed by an inverse-degenerate TOPC results in an AOPDP device that passes the incoming wave through only if its polarization matches the conditions in equation (14).

After describing the AOPDP operation concept we turn to estimating its performances. Specifically, we will try to estimate the 'finesse' of the AOPDP. Ideally, the AOPDP would pass through only one dot on the Poincaré sphere. However, in practice a small zone of the sphere will be passed.

Three elements determine the performance of the AOPDP—the TOPCs (specifically the MO rotators), the optical amplifier and the PR switch. We do not regard physical geometric accuracy as a factor since we assume that satisfactory accuracy is achievable.

In order to achieve good 'finesse' it is essential that we apply a highly non-linear function on the control signal, i.e. on the signal that is refracted off the PBS and is directed into

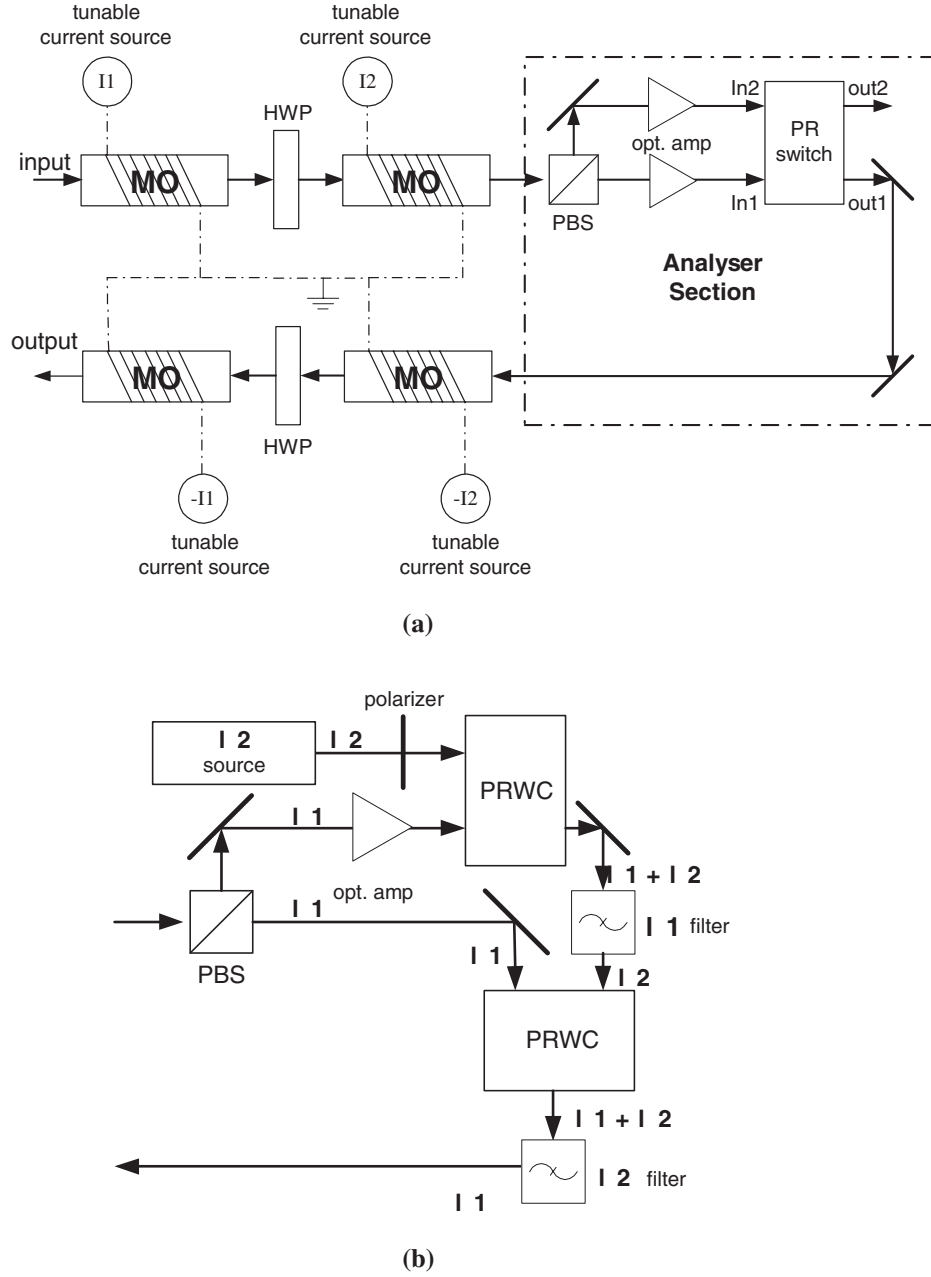


Figure 6. (a) Scheme of the AOPDP using a wave mixing PR switch. (b) Alternative analyser section implementation using the photo-refractive wavelength converter as switch.

In2 of the PR switch. This non-linear function practically separates the special situation where no signal or a very small signal is directed into In2 from the other situations where that signal's intensity varies from small fractions of the input signal to 100% of the input signal. The sharper and more precise is the separation, the higher is the 'finesse' that will be received. Since the PR switch does not apply such a strong non-linear response we place saturated optical amplifiers to achieve high non-linearity. Optical amplifiers can produce an order of 10 dB range of input power while remaining in saturated mode. Adding the amplifier non-linearity to the function of the PR switch can lead to a satisfactory non-linear function. If needed, two stages of amplification can be used, or other configurations may be suitable for that matter.

The PR switch also affects the system's extinction ratio (ER). In fact, the PR switch is the main component in determining the ER of the AOPDP. The ER of the PR switches is generally quite high (>20 dB) and has the potential to reach very high values.

We have performed a calculation of a possible ideal system of the AOPDP to get an idea of the AOPDP's performance. We assumed that a two-stage amplifier set-up was used and also assumed that an amplifier was placed before the PBS to enhance the input signal. The amplifiers' performances were assumed to be standard for saturation mode amplifiers. The amplifiers are assumed to be in saturation mode if their input signal is above -10 dB m. The amplifier output power level in saturation is 16 dB m and the amplifier gain is 25 dB when

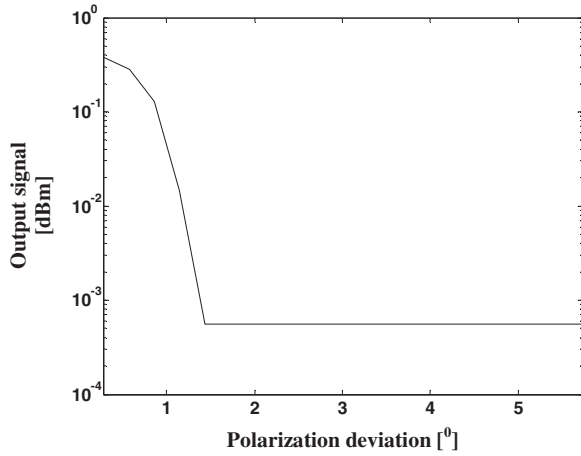


Figure 7. Calculation of the analyser section function. The output signal power is shown versus the polarization orientation deviation of the signal entering the analyser section off the PBS chosen axis.

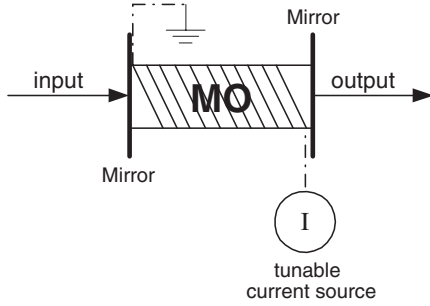


Figure 8. Scheme of the MOFP.

not in saturation. The PR switch performances are based on the switch in [8] when no noise is present. Figure 7 depicts the calculated function representing the analyser section. As one can see, deviations in the polarization orientation of the signal coming into the PBS of more than 1.5° cause a more than 30 dB decrease in the output signal's power. Such a strong non-linear function enables a resolution of approximately 120 pixels (dots) on the Poincaré sphere. The received 'finesse' can be improved by using different PR switches that produce greater non-linearity.

Finally, the TOPCs do not affect the device's 'finesse'. The MO rotators enable us to adjust the wave's polarization in resolution of the order of 0.1° . The MO resolution is much finer than the resolution imposed by the non-linear function of the analyser section (amplifier + PR switch) therefore the TOPCs do not affect the 'finesse'. The TOPCs can create some small deviations in the determination of the centre polarization (Poincaré dot) that is passed through. It is important to mention that the MO rotators might experience some drifting effects that should be compensated by changes in the applied currents.

6. Magneto-optic Fabry–Perot (MOFP) resonator

We now present a theoretical investigation of a FP resonator that contains a MO rotator in between its mirrors (see figure 8). Intuitively it is quite difficult to anticipate the effect of the MOFP resonator. However, it will be shown that the resonator produces enhancement of the MO effect.

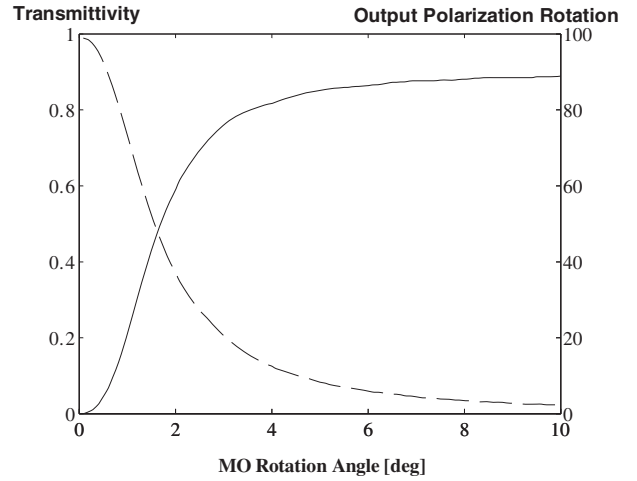


Figure 9. Normalized transmittivity of the MOFP and polarization of the output field versus MO rotating angle. Dashed curve represents transmittivity.

Consider an $\hat{x}$ polarized CW radiation that is directed into the MOFP. The field complex amplitude function at the exit of the MOFP, assuming that the resonance condition is fulfilled, is given by

$$\mathbf{E}_{\text{out}} = t^2(\cos \alpha + r^2 \cos 3\alpha + r^4 \cos 5\alpha + \dots)\hat{x} + t^2(\sin \alpha + r^2 \sin 3\alpha + r^4 \sin 5\alpha + \dots)\hat{y}, \quad (15)$$

where α is the rotation angle created by the MO medium in one pass and r, t are the amplitude reflection and transmission coefficients respectively. We assume that the FP is in resonance and that there is no internal resonance created between the MO medium and the FP mirrors.

We have calculated numerically the power transmitted into $\mathbf{E}_{\text{out}}$ and its polarization. The result is presented in figure 9. $R = r^2$ was chosen to be 0.95. One can see that 45° rotation of the output polarization is achieved with less than 1.6° polarization rotation induced by the MO rotator, and 10° rotation induced by the MO rotator produce 88.5° rotation of the output polarization. The last result shows that the MO resonator has created a significant effective increment of the MO effect. A similar yet different approach was suggested to increase the MO effect using a multi-layer MO structure [6]. Similar results were achieved.

Another interesting property of the MOFP is its frequency transmission function. In order to represent the frequency dependence we used the phase mismatch parameter that is given by

$$\Delta\psi(\lambda) = 4\pi nL \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right) = \frac{4\pi nL}{c} (f - f_0), \quad (16)$$

where c is the speed of light in vacuum, n is the refractive index of the FP medium, L is the resonator's length, λ_0, f_0 are a resonance wavelength and resonance frequency respectively ($\lambda_0 = \frac{2nL}{m}$, m —integer) and λ, f are the output wavelength and frequency respectively. Figure 10 shows the calculation of power transmittivity as function of the MO rotating angle and the phase mismatch of the FP resonator. Two interesting phenomena appear in this figure. The first is that the MOFP can be used as a tunable filter. Moreover, increasing the rotation

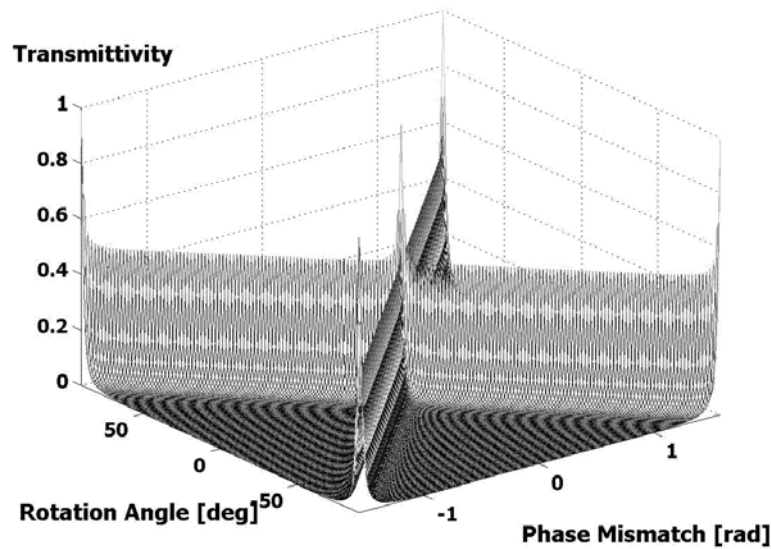


Figure 10. Normalized transmittivity of the MOFP as a function of the resonator phase mismatch and the MO rotation angle.

angle of the MO medium by δ degrees will shift the filter by the same amount in the $\Delta\psi$ axis. The second phenomenon is that the filtering function splits into two filtering functions. As one can see in figure 10, the function of the standard FP resonator that is received when the rotation angle of the MO medium is zero splits into two peaks when rotating angle starts to increase. The two peaks have identical amplitudes and are equally distant from the zero mismatch point ($\lambda = \lambda_0$). Their polarization is also identical except for the sign of the relative phase between the $\hat{x}$ and $\hat{y}$ components, i.e. one peak has a left rotating polarization while the other has a right rotating polarization. Note that when the rotating angle reaches 90° the peaks recombine and create the same frequency transmission function as in 0° rotation.

7. Conclusions

In this paper we have analysed and experimentally characterized the properties of MO materials. We have presented three different devices that take advantage of the MO effect. These devices are, generally speaking, polarization handling devices. The main characteristic of the devices is the simplicity of their design concept.

The main advantage of the TOPC is that it uses extremely simple feedback signals to perform polarization conversion. This may be attractive for use in different systems.

The AOPDP complexity is higher than that of the TOPC. However, it offers a very important capability of polarization filtering. The fact that we have designed an all-optical device makes the latter point even stronger.

Finally, we showed that a very simple device such as the MOFP can produce a more than one order of magnitude enhancement of the MO effect. The MOFP could significantly decrease the needed thickness of a MO medium for polarization rotating. It can also improve the switching time of all manner of switches that use the MO medium.

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